AuraData: Autonomous FinOps Intelligence

Agentic Cloud Cost Optimization & Financial Governance

Powered by Google Gemini 2.5 Flash

The Challenge: Cloud Chaos

Enterprise cloud spend is growing 20% year-over-year, yet 30% of that spend is wasted on idle, over-provisioned, or orphaned resources.

Why Costs Spiral?

Visibility Gaps

Finance teams lack real-time access to engineering infrastructure, creating a lag between spending and auditing.

Scalability Issues

Manual audits cannot keep up with thousands of microservices and ephemeral serverless functions.

The Solution: Agentic FinOps

- **Real-time Querying:** AI interacts directly with live databases.
- **Autonomous Reasoning:** Identifies waste patterns without human intervention.
- **Context-Aware Optimization:** Suggests right-sizing based on actual service utilization.

Core Capabilities

Secure Tool Use

Leverages Model Context Protocol (MCP) for restricted, read-only database access.

Governance

Automated policy enforcement prevents costly infrastructure "hallucinations."

C-Suite Reports

Translates complex technical data into concise executive summaries.

System Architecture

A multi-layered stack designed for high-speed agentic reasoning and data integrity.

The Need for Automation

AuraData vs Traditional Audits

Feature	Manual FinOps	AuraData Agent
Analysis Speed	2-5 Days	< 5 Seconds
Data Integrity	Prone to Human Error	Direct Tool Execution
Optimization ROI	Reactive	Proactive/Autonomous

Projected Impact

35%

Potential Monthly Savings

Driving Efficiency

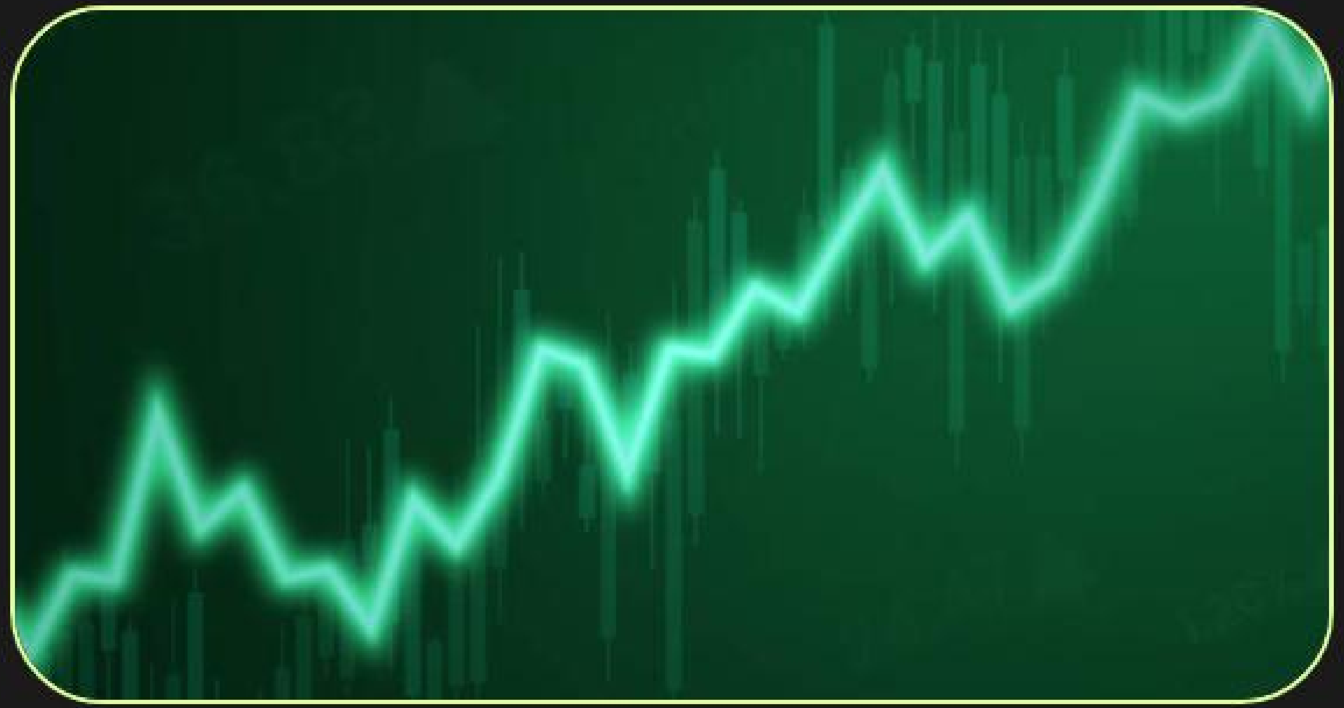
By automating the detection of idle EC2 instances and right-sizing Lambda functions, AuraData identifies significant savings in minutes that human auditors miss.

Enterprise Success Model

Engineering Team

Engineering

Automated resource reclamation for AWS workloads.



Finance

Zero-lag reporting for departmental cloud budgets.

Compliance

Rigid security guardrails via read-only Tool access.

Built on Gemini

AuraData leverages **Gemini 2.5 Flash** for ultra-low latency reasoning and advanced tool-calling, ensuring the agent remains fast, accurate, and cost-effective.

Ready for Scale.

GitHub Repository | YouTube Demo | Kaggle Writeup

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Image Sources

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